

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (Biotechnology) (Credit System) Degree Examinations**

January 2023

20BTC111 - NUTRACEUTICALS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain the nutritional components of bioactive peptides.	8	L*2	1	3,6
b)	Discuss the sources, deficiency, daily requirement and health benefits of fat soluble vitamins.	8	L2	1	3,6
c)	Write a note on soybean as functional foods.	4	L2	2	3,6
2. a)	Explain polyphenols, its types and health benefits.	10	L1	2	3,6
b)	Describe fish as specific functional food.	10	L2	2	3,6
3. a)	Write a note on polyunsaturated fatty acids.	8	L2	2	3,6
b)	Comment on iron as microminerals.	4	L2	2	3,6
c)	Elaborate on the soluble and insoluble dietary fibers.	8	L2	2	3,6
Unit – II					
4. a)	Define secondary metabolite and discuss on the extraction, purification and applications of alkaloids.	10	L2	3	3,4,6
b)	Explain the sources, extraction and applications of: i) glucosamine ii) chondroitin sulfate.	10	L2	3	3,4,6
5. a)	Explain the role of synbiotic for maintaining good health.	10	L2	4	3,6
b)	Write a note on bacteria as probiotics.	5	L2	4	3,6
c)	Explain algae as source of minerals.	5	L2	4	3,4,6
6. a)	Describe the extraction of antioxidants from algae and add a note on its applications.	10	L2	4	4,6
b)	Explain the production and extraction of prebiotics and probiotics.	10	L2	4	4,6
Unit – III					
7. a)	Describe the role of nutraceuticals with reference to hypercholesterolemia.	10	L1	5	3,6
b)	Define antioxidants and briefly explain the types.	4	L1	5	3,6
c)	Explain how antioxidants can be used as supplements in treatment of stress.	6	L2	5	3,6
8. a)	Discuss the role of nutraceuticals with reference to diabetic mellitus.	10	L2	5	3,6
b)	Explain the role of nutraceuticals in control of obesity.	10	L1	5	3,6

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
